

AERIS-TWIN

AI-Enabled Digital Engine Intelligence System for MALE UAVs

"The engine tells you it is about to fail - six hours before it does."

AERIS-TWIN is a digital twin that converts raw engine telemetry into a failure prediction, a remaining-useful-life countdown, and a specific action for the ground crew.

TEAM NAME

[Team Name]

PROBLEM STATEMENT ID

[PS ID]

PROBLEM STATEMENT TITLE

[PS Title]

SOLUTION NAME

AERIS-TWIN - Digital Engine Intelligence System

ROTAX 914 | FLAT-4 | 115 HP

CYL 1

154°C

CHT

CYL 2

231°C

CHT

TURBOCHARGER - FULL BOOST

CYL 3

151°C

CHT

CYL 4

157°C

CHT

DIGITAL TWIN - LIVE STATE

HEALTH INDEX

71%

RUL REMAINING

412 h

ANOMALY SCORE

0.62

ALERT: CYL 2 OVERHEAT - cooling duct restriction, CHT +74°C above baseline.

20 Hz

coupled physics twin

112 B / frame

20 Hz binary telemetry ~18 kbps

20-40 ms

measured LOS one-way latency

220-520 ms

SATCOM - measured, never hidden

PROBLEM | SOLUTION | NOVELTY

Problem Understanding & Proposed Solution

01 THE PROBLEM

- TAPAS BH-201: 8 years and INR 1,786 crore spent, project closed January 2024.
- Sensors worked. The intelligence layer did not exist - 10+ channels logged, never interpreted.
- Fixed 1,200 h overhaul interval: too early (wasted cost) or too late (catastrophic).
- Failure is never sudden. Rising CHT, drifting EGT and vibration always precede it - warning time today is zero.



02 OUR SOLUTION

- 1 Real-time digital twin**
20 Hz coupled physics model of the Rotax 914 across 10 live channels.
- 2 Multi-parameter anomaly detection**
Isolation Forest over all channels + FFT isolation of the 140 Hz bearing (BPFO) signature - not single-threshold alarms.
- 3 Weibull RUL + explainable diagnostics**
Hours-to-failure countdown; every alert names its contributing factors and confidence.
- 4 JARVIS conversational copilot**
The operator asks anything and gets an answer grounded in the live aircraft snapshot.

200 readings/sec -> 1 health index + 1 actionable advisory

03 NOVELTY - WHY THIS IS NOT ANOTHER DASHBOARD

PREDICTIVE, NOT REACTIVE

Computes the future state and time-to-failure, not just the present value.

EXPLAINABLE BY DESIGN

Confidence and contributing factors on every alert - built to defeat false-alarm distrust.

CAN-NATIVE FRONT END

Consumes the exact 8-byte scaled frames a real FADEC emits. Simulation to hardware = one transport swap.

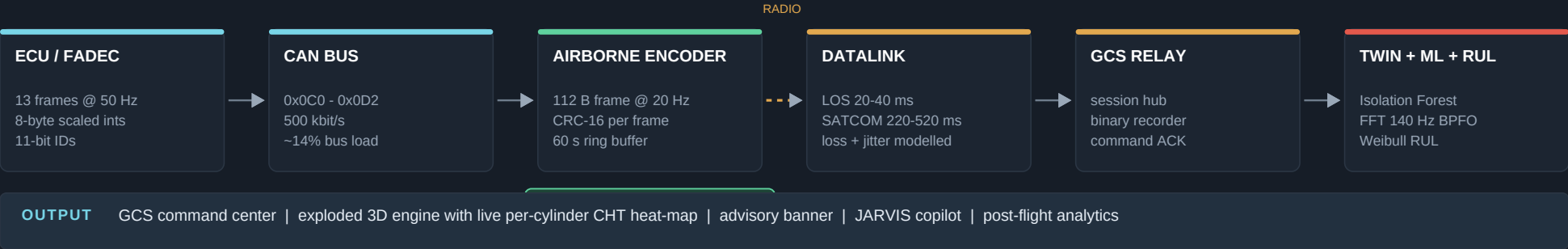
ENGINE-AGNOSTIC

Parameterised frame map and model: Rotax today, any piston or turboprop tomorrow.

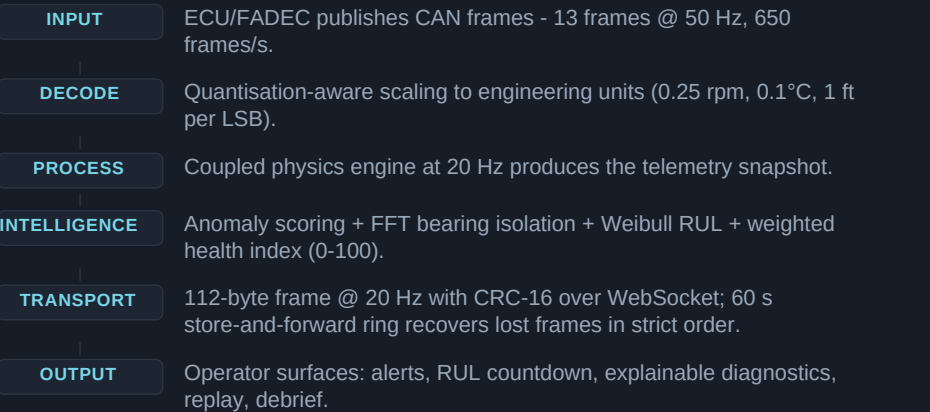
INPUT -> PROCESSING -> OUTPUT

Technical Architecture & Approach

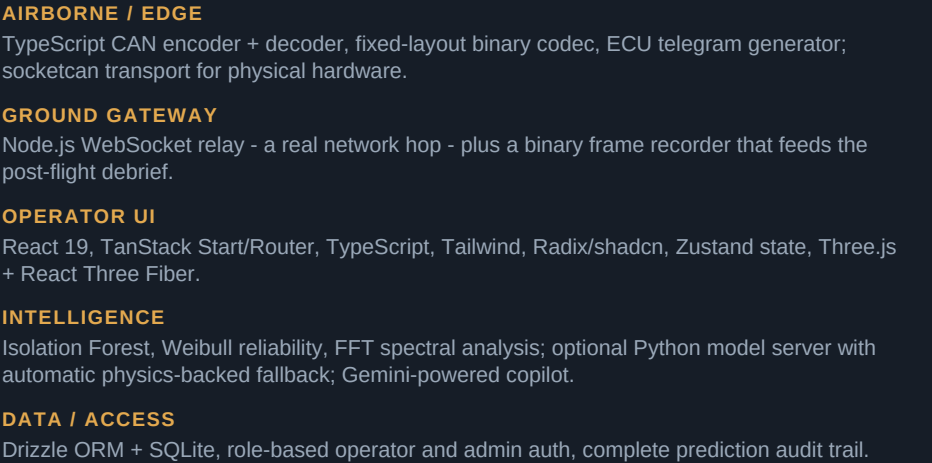
END-TO-END DATA PATH - SENSOR TO OPERATOR



01 DATA FLOW: INPUT -> OUTPUT

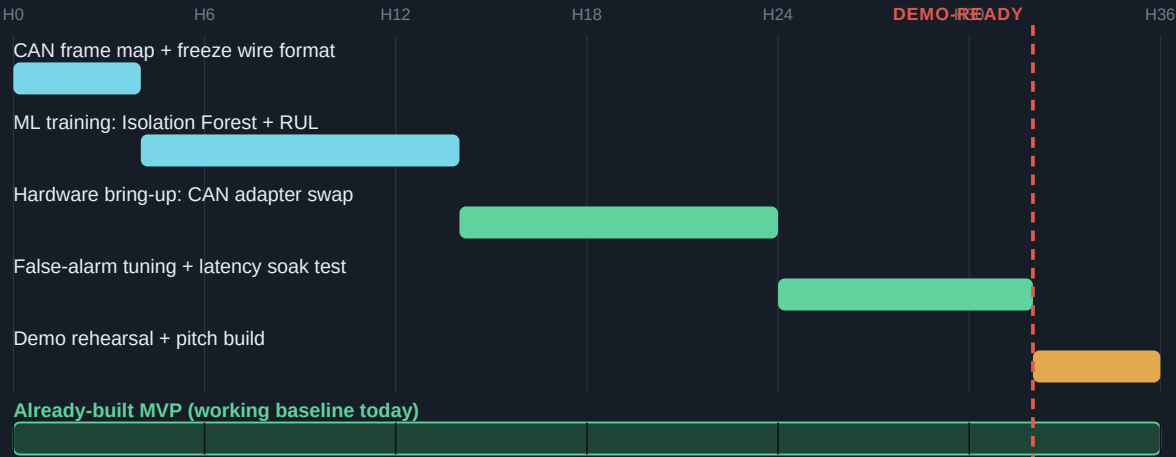


02 TECH STACK BY LAYER



Feasibility & Viability

01 36-HOUR BUILD PLAN



Team split: 2 frontend/3D | 2 backend/datalink | 1 ML | 1 hardware + telemetry. The 36 hours harden a system that already runs - they do not build it from zero.

ALREADY BUILT - THE 36-HOUR PROOF

- Three live surfaces: landing, airborne sim, GCS command center.
- Two-window datalink over a real WebSocket hop, with latency, loss and CRC measured live.
- CAN ingestion layer, anomaly scoring, RUL countdown, mission replay, post-flight analytics.

02 RESOURCES

- OPEN SOURCE**
React + Three.js, ws, Drizzle, scikit-learn (IsolationForest), SciPy (FFT), NumPy.
- DATASETS**
NASA C-MAPSS turbofan degradation, Rotax 914 published performance data, standard-atmosphere tables.
- HARDWARE**
USB-CAN adapter, Raspberry Pi-class edge node, engine test cell or a recorded ECU trace.
- APIS**
Gemini for the copilot; optional weather and airspace feeds. No licensed component required to demo.

03 SCALABILITY - 1 TO 1,000,000

1 AIRFRAME

Every byte already crosses a real socket; the twin is the same code path.

10 AIRFRAMES

Relay state is per-session (60 s ring) - add a session, not a service.

1,000 AIRFRAMES

Fixed 112 B/frame = ~1.6 GB per aircraft per flight-day. Shard sessions; storage is a rounding error.

1,000,000 USERS

The analytics and debrief layer is a read-only API over recorded frames - horizontally trivial.

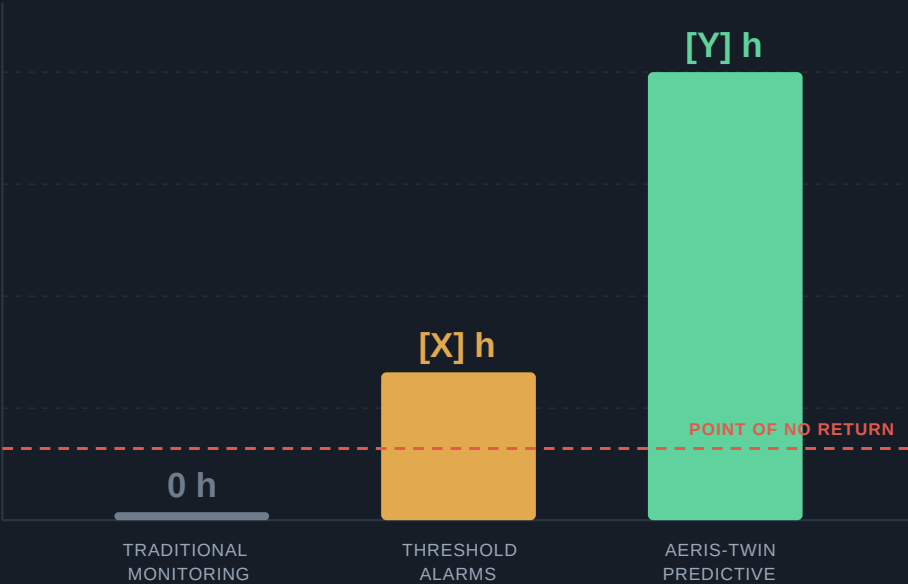
ZERO INFRASTRUCTURE

Edge inference on board delivers alerts with no ground segment at all.

USERS | QUANTIFIED OUTCOMES | EVIDENCE

Impact & Benefits

01 DETECTION LEAD TIME (HOURS)



Illustrative geometry - replace [X] / [Y] with your validated measurements.

02 WHO THIS SERVES

- ✓ **GROUND OPERATORS / MISSION ENGINEERS**
high-altitude endurance UAV fleets
- ✓ **MAINTENANCE PLANNERS + AIRWORTHINESS**
prediction evidence and a full audit trail
- ✓ **EVERY PISTON / TURBOPROP OPERATOR**
trainers, ISR, ag-drones, remote power generation

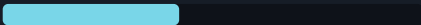
03 QUANTIFIED IMPACT

- Unplanned engine events**
target -[X]% per 1,000 flight hours [SRC]
- Avoided premature overhauls**
1,200 h fixed TBO -> condition-based, ~[X]% avoided [SRC]
- Asset protection**
one prevented engine + airframe loss = INR [X] crore
- Program risk**
TAPAS-class INR 1,786 crore - signature was diagnosable years earlier
- Operator workload**
200 readings/sec collapsed to 1 health index + 1 advisory

04 VALUE ACCRUED PER 100 FLIGHT HOURS

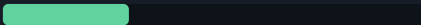
AVOIDED OVERHAULS

[X]%



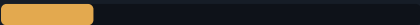
PREVENTED EVENTS

[X] events



REDUCED DOWNTIME

[X] h



NET COST AVOIDED

INR [X]



Research, Market & References

01 MARKET CONTEXT

- Aerospace predictive maintenance and digital twins growing at [X]% CAGR to \$[X]B by [year] [SRC].
- India's pull: indigenous MALE/HALE programs, HAL, private UAV OEMs and fleet sustainment under mandate to cut imports.
- Regulatory tailwind: condition-based maintenance approval requires demonstrated prediction accuracy - our audit trail produces exactly that evidence.

02 EXISTING SOLUTIONS vs OURS

- OEM / FADEC diagnostics**
Threshold display only, engine-locked. No prediction, no explanation.
- Enterprise CBM platforms**
Ground-only offline analytics at procurement-scale cost. No live twin.
- SCADA / telemetry loggers**
Record and plot data. Never interpret it.

OUR MOAT: prediction + explainability + CAN-native + edge-capable

03 COMPETITIVE POSITION



Deployable in a weekend, not a procurement cycle.

04 REFERENCES

[1] DRDO/ADE TAPAS BH-201 program status and performance data - [SRC]

[2] Rotax 914 overhaul / operator manual - 1,200 h TBO, CHT & EGT limits - [SRC]

[3] NASA C-MAPSS turbofan degradation dataset - [SRC]

[4] Liu et al., Isolation Forest (2008) - [SRC]

[5] Weibull reliability and RUL modelling reference - [SRC]

[6] Bearing outer-race fault frequency (BPFO) derivation - [SRC]

[7] SATCOM link latency and link-budget analysis - [SRC]

[8] UAV predictive-maintenance market size - [SRC]